Electrical Power and Energy Systems 134 (2022) 107399

Contents lists available at ScienceDirect

International Journal of Electrical Power and Energy Systems

journal homepage: www.elsevier.com/locate/ijepes

A real-time hierarchical framework for fault detection, classification, and location in power systems using PMUs data and deep learning

Mohammad Reza Shadi, Mohammad-Taghi Ameli *, Sasan Azad

Department of Electrical Engineering, Shahid Beheshti University, Tehran, Iran

ARTICLE INFO

Keywords:
Frequency Disturbance Event (FDE)
Phasor Measurement Unit (PMU)
Deep Learning (DL)
Event classification
Event locating
Fault Location

ABSTRACT

Frequency Disturbance Events (FDEs) occur due to various events such as Generator Trip (GT), Line Outage (LO), and Load Disconnection (LD), which affect the stability of power systems. Depending on disturbance severity, it has small to severe effects on an integrated system's performance. Accurate and fast detection and classification of the event and its location are crucial in monitoring resource adequacy, preventing economic losses and blackouts. This paper, focusing on an online cutting-edge hierarchical methodology, first detects the event, then identifies its classification. Eventually, the exact location of FDE is found based on Phasor Measurement Units (PMUs) data. Compared to other counterparts, which focused only on classifying or locating events separately, in this paper, both aims are achieved using the suggested novel hierarchical framework. This research employs Deep Learning (DL) advances to develop a Recurrent Neural Network (RNN) model and a Long Short-Term Memory (LSTM) model to distinguish and locate FDEs with significant accuracy. In the research, only a few time-series of Rate-Of-Change-Of-Frequency (ROCOF) received from a limited number of PMUs is used as the DL algorithm's inputs. This hierarchical methodology is tested in New England 39-bus, IEEE 14 bus systems, and modified IEEE 118-bus system. The evaluation results demonstrate the potential application of the proposed models to detect and classify FDEs compared to conventional algorithms and the frequency-based DL model. The proposed models have achieved significant classification accuracy. Simultaneously, the models locate the events with competitive performance in terms of precision, recall, accuracy, and F1-score.

© 2017 Elsevier Inc. All rights reserved.

1. Introduction

Power systems' ability to maintain stability and ensure a continuous and uninterrupted supply of customers' demands are of great importance. With the expansion of power grids in large geographical areas and the creation of larger and more complex grids, the probability of encountering a variety of faults and disturbances has increased. The behavior and hardness of disturbances are different in every operating condition of the power system and their impacts depend on the event's type, location, and duration [1]. Small disturbances occur in load changes, and large disturbances include Generator Trip (GT) [2], Line Outage (LO) [3], and Load Disconnection (LD) [4]. Large disturbances significantly affect the entire network and have critical consequences such as widespread blackout and considerable economic loss, such as the Northeast blackout of 2003, which led to power outages for nearly 50 million Americans and cost between $7 and $10 billion [5].

Major Frequency Disturbance Events (FDEs) have occurred in abundance in most countries and networks, such as blackout in Brazil (1996) [6], Iran and Italy (2003) [7], and Greece (2006) [8], which caused serious problems including partial or wide-area blackout, splitting grids into several islands, and repair and restoration costs. Therefore, the detection and classification of every FDE and its exact location must be done online with very high accuracy. Eventually, by generating an alarm signal, the control center can quickly implement appropriate control commands according to the event's type and location to ensure the system's security and stability.

The detection of frequency disturbances has become much more accessible due to the widespread utilization of Phasor Measurement Unit (PMU) in power systems. Since PMUs measure the synchronized phasors of voltage, current, frequency, and Rate-Of-Change-Of-Frequency (ROCOF) signals in real-time, they provide the Phasor Data Concentrator (PDC) and control center with the possibility of monitoring, protecting, and controlling [9–11]. Compared to traditional Supervisory Control and Data Acquisition (SCADA) measurements, which are

* Corresponding author.
E-mail address: m_ameli@sbu.ac.ir (M.-T. Ameli).

https://doi.org/10.1016/j.ijepes.2021.107399
Received 28 February 2021; Received in revised form 2 June 2021; Accepted 11 July 2021
Available online 28 July 2021
0142-0615/© 2021 Elsevier Ltd. All rights reserved.

sampled data every 2 to 4 s, PMU measurements have a higher sampling rate and can collect mentioned signals between 10 and 60 (depending on system frequency) samples per second. They are also superior to SCADA measurements in terms of quality and speed and can capture FDEs with high precision [4].

In recent years, PMUs have been widely employed for various applications such as voltage stability assessment [12], event detection [13], locating disturbance source [14], and detection and classification of disturbances [15]. Continuous monitoring of this amount of data gathered by PMUs and tracking events will be almost impossible for operators in real-time. To address this issue as well as reduce the computation burden, intelligent algorithms like Machine Learning (ML) and Deep Learning (DL) can play a crucial role, which have shown the promising ability to resolve complicated nonlinear problems [4].

The detection, classification, and Localization of FDEs are categorized into data-driven and model-based approaches in the literature [16]. Recent advances in the use of intelligent algorithms have paved the way for data-driven approaches in the electricity industry [17,18]. New technologies such as power electronics and renewable energy sources in power grids and smart grids have made the system more nonlinear. Given that data-driven approaches are not reliant on the system structure, they can be beneficial in identifying FDEs for highly nonlinear systems. Some data-driven approaches have been proposed to load forecasting based on reinforcement learning [19], detection FDEs using Convolutional Neural Network (CNN) [20], fault classification based on bidirectional Long Short-Term Memory (LSTM) [21], and detection of false data injection attacks in an AC smart island using DL [22].

Recurrent networks are robust deep learning models with remarkable advantages compared to traditional neural networks on sequence data. For many feature sequence extraction, time series classification, and regression applications, Recurrent Neural Network (RNN) and LSTM demonstrate excellent performance and efficiency. They can handle data sequences because a long multidimensional feature pattern is modeled to classify a particular case in its correct class or map the whole sequence into a predicted scalar in regression issues. In addition, extensive experimental results show that RNN and LSTM can better separate types of events in power systems and provide more promising event classification performance. The LSTM model's superiority in order to diagnose fault from time series proved in [23]. In [24], the LSTM network is also employed for feature extraction for photovoltaic (PV) array fault diagnosis, which has proved the model's efficiency, especially on noisy data. To the authors' knowledge, RNN and LSTM models were not investigated already for classification and location FDEs in multi-machine power systems.

There are many works related to disturbance or event detection and classification have been reported. In [25], a novel mathematical methodology is suggested to identify multiple LOs in power systems based on current phasors of PMUs. Although this method can detect multiple LOs and the outage of parallel lines well, this method only focuses on LO detection. Similarly, a methodology for online detection only fault-induced LO using the concept of generator inertia is presented in [26] by dividing the whole network into different protection zone. A new index, named "gain in momentum", is utilized to identify tripped zone. Ref. [27] has investigated a novel method based on the pattern recognition approach for identification disturbances. This method uses the Wide Area Measurement System (WAMS) data, including voltage, current, active power, frequency, and phase angle to be able to detect events by extracting a feature. Therefore, a large amount of data collected is required to identify disturbances in the power system. In [2], using a decision tree algorithm and frequency data collected from Frequency Disturbance Recorders (FDRs), only GT events' identification is discussed. Ref. [28] proposed a new scheme Cluster-based Sparse Coding (CSC) algorithm to detect multiple events, including GT, LO, and LD, in large-scale power systems. CSC uses time-series frequency signals from all buses to find clusters of buses that behave similarly, notwithstanding the event type.

GT, LO, LD, and fault events cause a sudden and fast change in the entire network's frequency signal, affecting voltage and rotor angle stability. Hence the frequency signal is one of the important variables to identify these FDEs. In addition to frequency signal, recent studies have used ROCOF signal in various applications. Although the use of either of these two signals provides satisfactory results, the ROCOF signal demonstrates the frequency changes more obviously under disturbance circumstances [9,20].

Most of the existing works related to FDEs concentrate on only a few methods: (1) event identification, (2) event identification and classification event, and (3) event detection and location. Also, in these studies, only limited events such as LO or GT have been investigated. Consequently, a robust comprehensive work is required to address these issues.

This paper aims to develop a cutting-edge hierarchical comprehensive algorithm for accurate detection, classification, and location FDEs in power systems based on RNN and PMU measurements. The work presented in this paper uses the only ROCOF time series received from network PMUs. Since the ROCOF signal is a welcome indicator for identifying FDEs, this paper utilizes an RNN model using the ROCOF time series to develop a reliable and accurate model to detect GT, LO, LD, and fault events. The fault in the paper refers to the 3-phase fault on the transmission lines as the most severe type. Respecting the number of PMUs installed in power grids is low, a limited number of PMU has been employed in the work. The performance of the proposed model is compared with the LSTM model. Using the online hierarchical framework, the network operator obtains full knowledge about FDE and the current state. Therefore, triggering system responses or proper immediate actions can be taken to prevent the event's development and the occurrence of subsequent trips and blackout. Table 1 compares and provides a better picture of the differences between the proposed method and other state-of-art methods for FDEs identification.

The salient contribution of the proposed framework compared to previous related works can be itemized as follows:

- A cutting-edge hierarchical framework is proposed to detect, classify, and locate the FDEs in the system. The classification and location FDEs in the proposed methodology is performed more accurately compared to counterparts.
- As a comprehensive approach, all FDEs affecting network performance, including GT, LO, LD, and fault, are evaluated. After detecting the occurrence of an FDE, the proposed model classifies and locates it in real-time.
- Very little ROCOF data is used to identify the various dimensions of an FDE. A limited number of PMUs are also employed to collect ROCOF time series. Hence, it has less computational complexity than other data-driven approaches, and it will be easier to supervise and store this small amount of information.
- High-precision RNN and LSTM models are employed online to recognize, classify and locate FDEs in real-time.

The rest of this paper is structured as follows. Section 2 demonstrates the proposed hierarchical framework and data acquisition from PMUs. Numerical outcomes, performance evaluation and accuracy of the suggested approach is carried out in Section 3.

2. The proposed methodology

2.1. Data acquisition from PMUs

Due to unpredictable load demand changes, the system frequency is usually not in equilibrium, and with increasing and decreasing load demand, the frequency decreases and increases, respectively. When an FDE event occurs in the system, regardless of its location and local voltage and current values, the whole system's frequency magnitude is affected. According to Eq. (1), frequency fluctuations directly influence

Table 1
Comparison of the state-of-art methods for FDEs identification.

Reference	[4]	[17]	[20]	[29]	[30]	[31]	This Paper
Detection FDEs	—	✓	✓	✓	✓	✓	✓
Classification FDEs	✓	—	✓	—	—	—	✓
Locating FDEs	—	✓	—	✓	✓	✓	✓
Data-driven Approach	✓	✓	✓	—	✓	✓	✓
Model-based Approach	—	—	—	✓	—	—	—
Class of Events	GT, LO, LD, Inter-area Oscillation	—	GT, LD	—	—	—	GT, LO, LD, Fault
Model	CNN	K-mean Clustering	CNN	New Index (the Sum of Squared Residuals)	Least Squares	New Index (Sharpness Index)	RNN, LSTM
Performance Metric	Classification Accuracy	Location Accuracy	Classification Accuracy, Precision, Recall	Estimation Error	Comparison of the method with some experimental cases	Detection Accuracy, Comparison of the method with some experimental cases	Classification Accuracy, Location Accuracy, Precision, Recall, F1-score
Input Variable	Frequency	Voltage, Frequency	ROCOF, Relative Angle Shift	Voltage, Current	Frequency	the Energy of Machine's Rotational Speed	ROCOF
Dataset	Brazil Grid	Power System in Southwestern Parts of the U.S.	the U.S. Eastern Interconnection	New England 39-bus system, IEEE 118 bus system	Power System in Southwestern Parts of the U.S.	New England 39-bus system, IEEE 68 bus system	IEEE 14 bus system, New England 39-bus system, modified IEEE 118-bus system

the rotation speed of the generator rotor [1].

$$f = \frac{p \times n}{60} \tag{1}$$

where f, p, and n are the system frequency, the number of generator poles, and synchronous speed, respectively. Hence, the occurrence of FDEs causes quick variations in system frequency and rotation speed of generators. So in this paper, PMUs will be installed at busbars connected to generators. Considering the ROCOF signal illustrates these variations more obviously than the frequency signal, the paper uses the ROCOF signal collected by PDC. The ROCOF equation is as follows:

$$ROCOF_{t(k)} = \frac{F_{t(k-\tau+1)} - F_{t(k)}}{t(k-\tau+1) - t(k)} \tag{2}$$

where the value of ROCOF at timestamp k is indicated by $ROCOF_{t(k)}$; $F_{t(k)}$ and τ are the frequency at timestamp k and the time interval, respectively. The PMU data sampling rate is set at 60 Hz. The location of the PMUs in the New England 39-bus system is shown in Fig. 1. Only 5 PMUs are used in the network, so the cost of installing and using PMUs for the recommended framework is meager, as well as data storage costs. In the paper, PMUs data is employed to develop the proposed models. However, in reality, FDR or Remote Terminal Unit (RTU) data can also be used to train models.

A frequency control mechanism is used in an integrated system to recover the system frequency when events happen. The primary control or governors are employed to restore the balance between generation capacity and consumer demand against small frequency deviations. If an event such as GT or LO makes a relatively constant frequency change in the network, then secondary and third frequency control is applied to bring the frequency to its nominal value [32].

Hence, in the GT event, the frequency immediately drops across the system. With the intervention of control mechanisms, the rest of the network generators try to compensate for the generation shortage and increase the frequency to equilibrium by increasing the production capacity (Fig. 2.a). In contrast, with the occurrence of LD, the frequency increases. In this case, after a period of transition, the generators reduce their output power to strike a balance between production and consumption (Fig. 2b). In the case of LO events and 3-phase faults on lines, network generators' behavior will vary depending on these events' location. Because LO can either cause one or more LDs or a GT, or this event can change the load flow in the system. For example, if the line between busbars 16 and 19 goes out (Fig. 1), G04, G05, and one load are isolated from the grid and form an island. Therefore, depending on LO and fault events' location, the frequency value can encounter different fluctuations (Fig. 2c, Fig. 2d).

Fig. 2 illustrates a specific event representing GT, LD, LO, and fault event that the ROCOF signal values obtained from 5 installed PMUs in the New England 39-bus system. As can be seen, each of the events presents a distinct behavior that facilitates events' classification. However, it will not be possible for humans to track them in real-time. Moreover, in each class, most events perform identical behavior, so it will be challenging to pinpoint each class's location. DL is applied in the paper to address these issues.

2.2. RNN and LSTM model

DL has emerged as an advanced subfield of ML algorithms, enhancing data-driven approaches' performance by generating a large hypothesis space. To accurately classify and locate events in real-time, this paper develops an RNN model and then compares the model's performance with another designed LSTM model.

A. Recurrent neural networks model

RNN is a superior family of artificial neural networks fitting for learning representations of time-series or sequential data for prediction and classification. Besides the feed-forward part of RNN, its memory distinguishes this algorithm from other data-driven methods so that its outputs depend on previous elements within the sequence. Thus, this unique feature enables better RNN performance for time-series tracking. The RNN model is described by Eqs. (3) and (4).

$$h_t = f(W_{xh}x_t + W_{hh}h_{t-1} + b_h) \tag{3}$$

$$y_t = W_{hy}h_t + b_y \tag{4}$$

where h_t, x_t, and y_t donate the state, input, and output vector at time t, respectively. W_{xh}, W_{hh}, W_{hy}, b_h, and b_y denote the input weight matrix, the hidden-to-hidden state weight matrix, the output weight matrix, the input bias matrix, and the output bias matrix, respectively. Also,

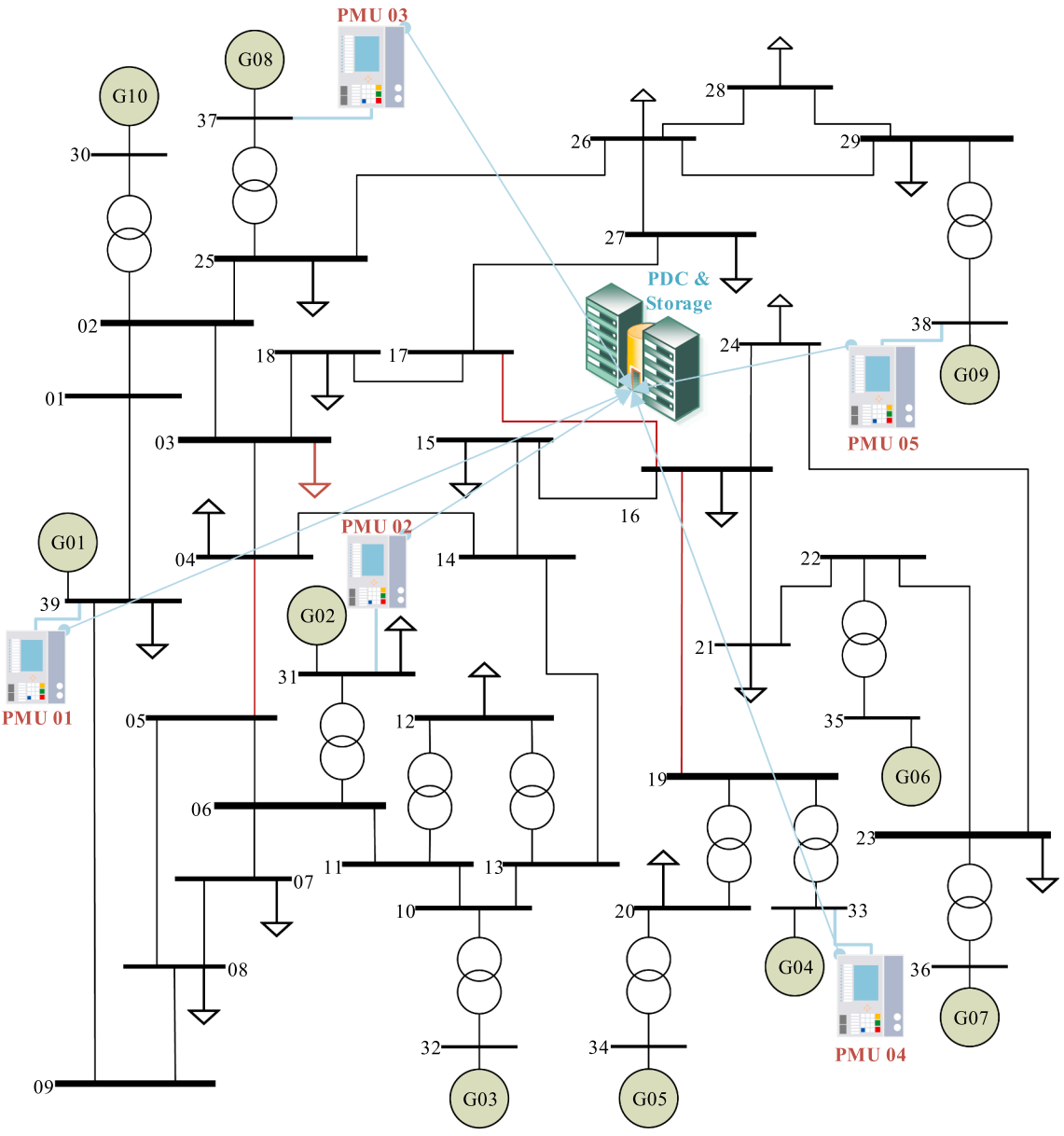


Fig. 1. Single-line diagram of the New England 39-bus test system.

nonlinear function f is activation function like ReLU or the hyperbolic tangent. Initially, h_{t-1} is typically considered to be equal to the zero vector. According to Eq. (4), the updated state vector has a direct effect on the output.

All weights matrices (W_{xh}, W_{hh}, W_{hy}) and all biases matrices (b_h, b_y) are the model parameters, requiring to be learned during training. The structure of an RNN and the chain representation of the RNN over time are shown in Fig. 3.

B. *Long-short term memory model*

The most important idea behind the LSTM model is called cell state. These networks are a particular type of recursive neural networks, learning long-term network dependencies. LSTM updates its temporal memory c_t (cell state) at each time-step according to the x_t input and then forms a temporal feature vector h_t. The LSTM model is described by Eqs. (5)–(10).

$$f_t = \sigma(W_{if}x_t + W_{hf}h_{t-1} + b_f) \tag{5}$$

$$i_t = \sigma(W_{ii}x_t + W_{hi}h_{t-1} + b_i) \tag{6}$$

$$g_t = tanh(W_{ig}x_t + W_{hg}h_{t-1} + b_g) \tag{7}$$

$$o_t = \sigma(W_{io}x_t + W_{ho}h_{t-1} + b_o) \tag{8}$$

$$c_t = f_t c_{t-1} + g_t' c_t \tag{9}$$

$$h_t = o_t tanh(c_t) \tag{10}$$

where f_t, or "forget gate," restricts the amount of information to be removed from c_t. This gate determines whether the memory information is used from the previous time-step or not. σ represents the sigmoid activation function. This layer's output is a number between zero and one, which indicates how much of the input should be sent to the output. The variable i_t is the "update gate," determining whether new information should be utilized at the current time-step and to what extent. It is also necessary to update the memory at each step, which is done by g_t. o_t is also the "output gate," deciding how much information from the previous time-step is transferred to the next time-step with the current

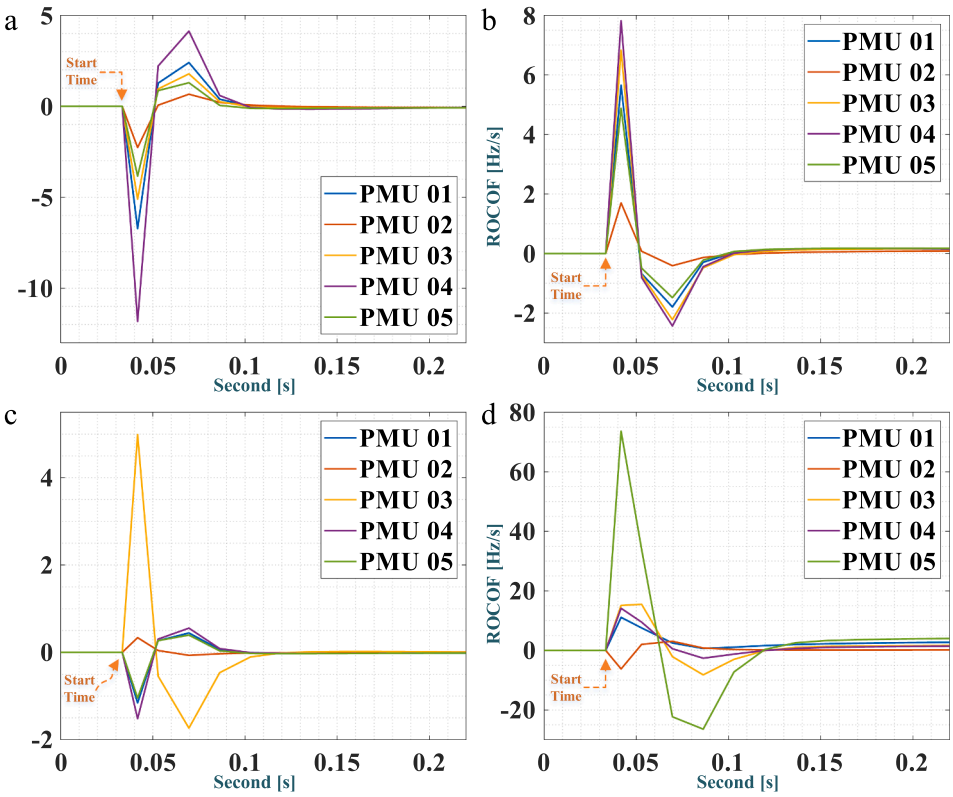


Fig. 2. Samples of ROCOF time series of each FDE in New England 39-bus test system; (a) Trip of G10; (b) Disconnection of Load 03 (located at Bus 03); (c) Outage of Line 04 - 05 (located between Bus 04 and Bus 05); (d) 3-phase fault on Line 16–17 (located between Bus 16 and Bus 17).

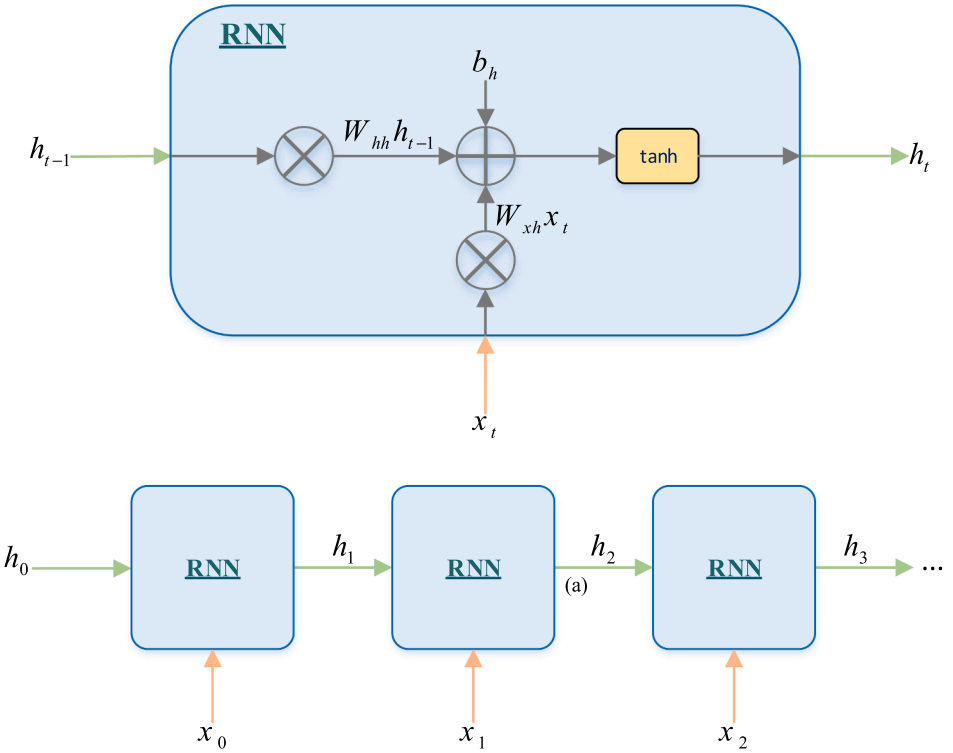


Fig. 3. Illustration of an RNN architecture.

time-step information. W_{if}, W_{hf}, W_{ii}, W_{hi}, W_{ig}, W_{hg}, W_{io}, and W_{ho} represent the LSTM weights for each gate. b_f, b_i, b_g, and b_o are also the biases for each cell. The structure of an LSTM and the chain representation of it over time are shown in Fig. 4.

C. *Description of the layers in RNN and LSTM*

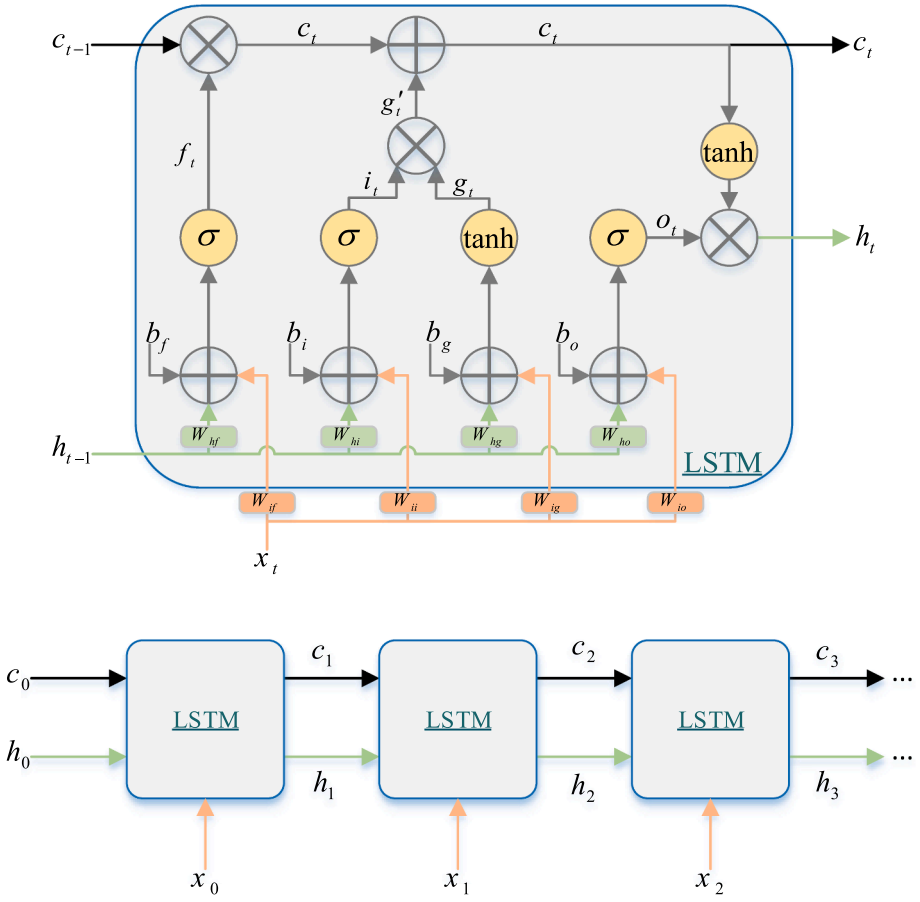


Fig. 4. Illustration of an LSTM architecture.

Each neural network consists of various successive layers. How the RNN and LSTM layers run is explained. In addition to these layers, a typical DL model has other layers, including the activation layer, the drop-out layer, the batch normalization (BN) layer, and the flatten layer. For classification schemes, dense layers and the softmax layer are commonly employed in the last step. The function of each of these layers is as follows:

***Activation layer*:** Using this nonlinear function helps the model learn intricate patterns in the data. The function receives the output signal from the previous cell and converts it to a form to be accepted as the next cell's input. The most popular activation functions are Rectified Linear Unit (ReLU) (11), sigmoid (12), and hyperbolic tangent (13) [33].

$$\alpha_{RELU}(x) = max(0, x) \tag{11}$$

$$\alpha_{sigmoid}(x) = \frac{1}{1 + e^{-x}} \tag{12}$$

$$\alpha_{tanh}(x) = tanh(x) = \frac{e^{x} - e^{-x}}{e^{x} + e^{-x}} \tag{13}$$

***Drop-out layer*:** Dropout is a powerful regularization strategy to avoid overfitting in DL, which during training, some layer outputs layer are randomly disconnected. It is eliminated neurons with a probability of p or maintained with a probability of $1 - p$.

***Batch normalization layer*:** BN can be considered as a method to develop the performance and training speed of DL. BN concentrates on distribution and normalizing input data to each layer, making the zero mean of data and unit variance. It is more effective than the dropout for defeating overfitting [34].

$$y_i(x) = \gamma \frac{x_i - \mu_x}{\sqrt{\sigma_x^2 + \varepsilon}} + \beta \tag{14}$$

where $\gamma$and β denote the scaling and shifting parameters, respectively; $\mu_x$and $\sigma_x^2$are also the mean and variance amounts of input x_i. $\varepsilon$is an adequately tiny positive parameter.

Flatten layer: Flatten layer is proposed to covert features into a 1-dimensional array, which can be used as inputs of the dense layer. So it is always applied before the dense layer.

***Dense layer*:** It is the most common layer in neural networks, known as the fully connected layer, in which all the neurons are connected to all the previous layer neurons. This layer is regularly utilized in the last stages of neural networks, which turns features into a space that makes them easier to classify.

***Softmax layer*:** It's a fully connected layer with the activation function softmax, which is often used as the neural network's last activation function. The layer has an equal number of neurons as the final classification, and its value determines the probability that the input data belongs to each of the classes. For the vector $x$with length j, the value of softmax is calculated by Eq. (15), where i is the vector index, $i = 1, 2, \ldots, j$[35].

$$S_i = \frac{e^{x_i}}{\sum_{i=1}^{j} e^{x_i}} \tag{15}$$

D. *Structure of the proposed RNN and LSTM models*

The architecture of the proposed both RNN and LSTM models is presented in Fig. 5. The two proposed models have the same structures, and the only difference is in the RNN and LSTM layers. Fig. 5a illustrates

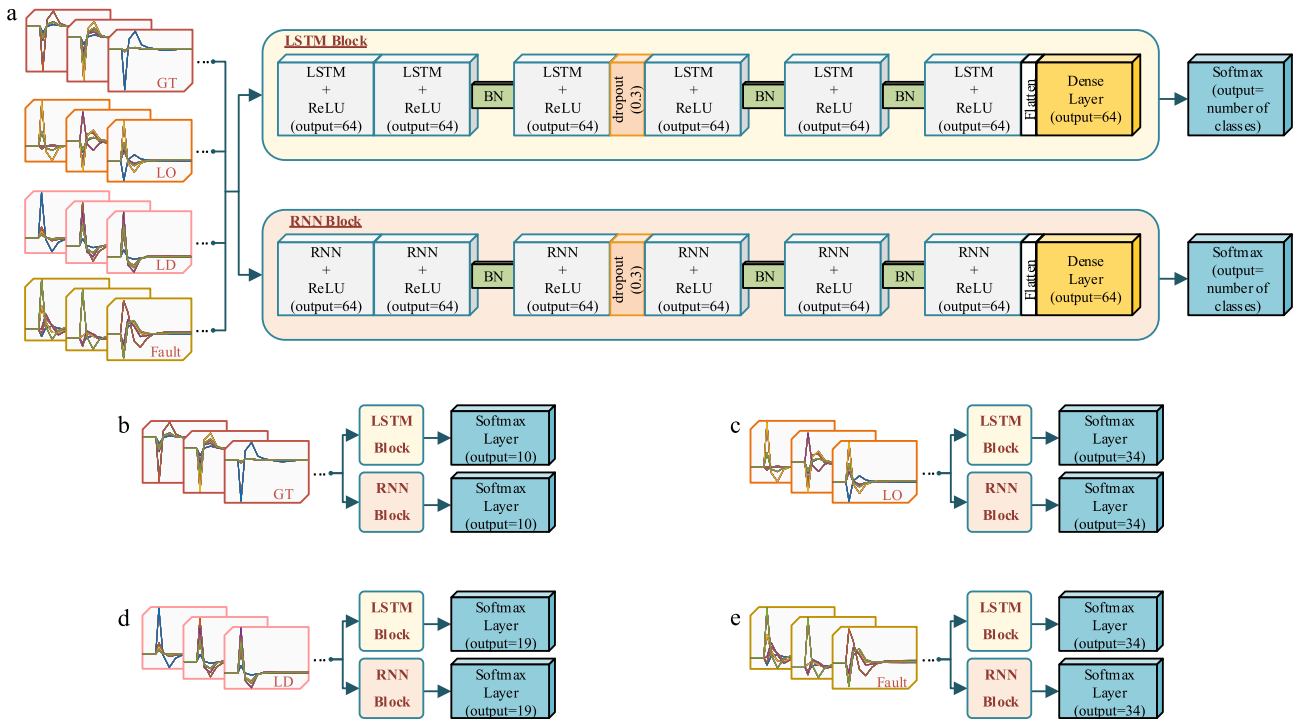


Fig. 5. Structure of the proposed LSTM and RNN models; (a) Event Classification; (b) GT location; (c) LO location; (d) LD location; (e) Fault location.

both models' structure for FDE type detection, which catch GT, LO, LD, and fault events as input for training. The frameworks of Fig. 5b, Fig. 5c, Fig. 5d, and Fig. 5e are applied to recognize the location of the GT, LO, LD, and fault, respectively, and each model handles related data to distinguish the location of the event. Also, due to the various number of expected locations in every event, the totality of outputs of the final softmax layer will equal the whole classes in each event. For instance, ten candidate generators exist in the New England 39-bus system as the GT event location, so the softmax layer in Fig. 5b generates ten possible quantities for the event and predicts the event location accordingly.

***Loss function*:** By comparing the predicted value of the model and the expected target value, determines whether the model is desirable or not using the definition of a penalty value. Therefore, the optimizer updates the model parameters by minimizing the loss function to increase the model's accuracy in predicting and classification. The sparse_categorical_crossentropy loss function has been selected in this study, which is applied by default for multiclass classification issues. To tuning the parameter and hyperparameters values of the proposed models, the values presented in Table 2 have been applied.

***Overfitting*:** This means that the trained model has worse performance and accuracy on new data over the training dataset. In each of the proposed RNN and LSTM models, there are approximately more than 50,000 and 190,000 parameters, respectively, whose values must be well trained to defeat overfitting. This study uses both BN [34] and dropout layers [36], which play a crucial role in mitigating overfitting. Only one dropout is employed in the middle of the proposed network according to need and strategy to prevent overfitting and remove only a few neurons and improve the model to train better.

All models used to classify and locate FDEs are trained offline by simulated events. During offline training of models, events are examined in different network operating situations to build the training database. Once the models are well trained, they can be employed to diagnose pre-trained and unforeseen FDEs online.

Table 2
Details of the parameters applied in the proposed models.

Hyperparameters	Selection
Optimizer	Adam
Loss Function	sparse_categorical_crossentropy
Learning Rate	0.001
Weight Decay	10^{-6}
Momentum	0.9
Training Batch Size	32

2.3. Proposed hierarchical methodology to identify FDEs

The novel proposed hierarchical framework for N-1 contingencies enables real-time evaluation of FDEs in power systems. The methodology consists of three crucial stages: detection, classification, and determining the FDEs location, shown in Fig. 6.

According to the North American Reliability Council (NERC), the minimum frequency criterion for detection FDEs is stated; (1) the frequency decreases by 0.45 Hz or increases by 1 Hz compared to the system's nominal frequency. (2) ROCOF exceeds 0.0124 Hz/s. In [37], a minimum of 0.05 Hz/s was employed as a disturbance detection criterion. Hence, the paper detects FDEs when ROCOF was passing the predetermined threshold value. Real-time ROCOF data collected from PMUs are entered into the proposed hierarchical framework. In the first stage, if the ROCOF values exceed 0.05 Hz/s, the PDC accumulates 15 samples of ROCOF signals, including 3 pre-disturbance and 12 post-disturbance data, from every installed PMU. This collected data are employed as the input of pre-trained RNN or LSTM model to distinguish the type of event. In the second stage, the FDE type is identified by the models whose output is GT, LO, LD, or fault (decision making 1 in Fig. 6).

In the last stage, depending on the type of FDE being detected in the second stage, the related pre-trained model will locate that event. Eventually, the location of the FDE is determined in decision making 2 in Fig. 6. Subsequently, depending on the FDE type and location, associated alarm signals will be sent to the system operator to have a complete sense and awareness of the network situation and take appropriate actions and control commands following the disturbance. Hence, in the quickest possible time, it is possible to respond to any FDE and prevent the spread of disturbance, local and global blackouts, as well as

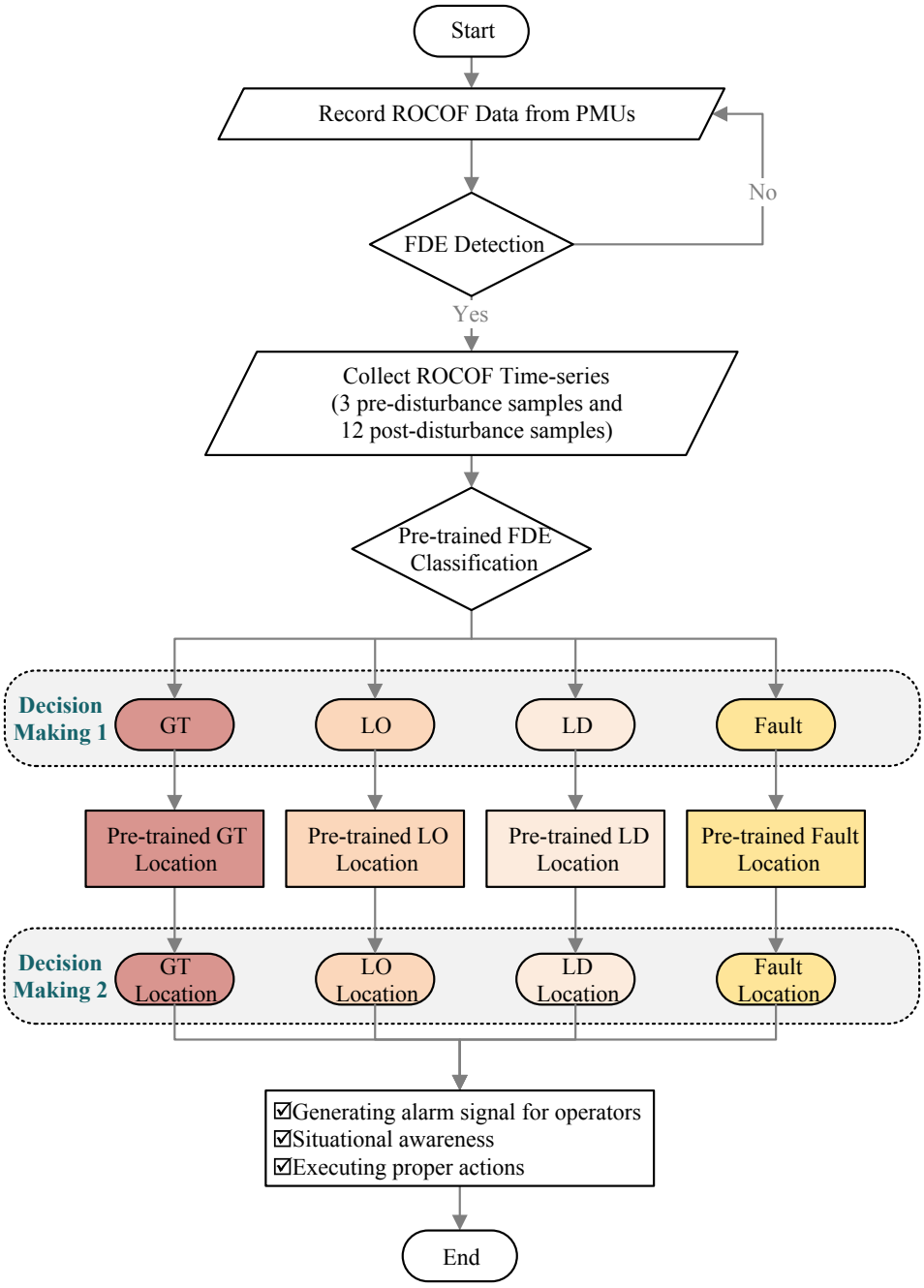


Fig. 6. Proposed hierarchical methodology for detection, classification, and location FDEs.

economic losses.

Pre-trained models are worked as a module in the PDC center. This module comprises a pre-trained model for detecting FDEs classes and four pre-trained models for locating GT, LO, LD, and fault events.

In using this framework, the model's accuracy in the second stage plays a vital role in the final decision to locate the FDEs. However, the proposed models and the ROCOF signal selected help to achieve remarkably high accuracy in this stage. If the model in the second stage misdiagnoses the class of FDE, not only the proposed model but also other existing works (such as [38,39]) will fail to achieve the objectives of the problem.

3. Numerical evaluation

Models training is conducted by programming in Python 3.8.3 using Keras on a personal computer with 16 GB of RAM and a 2.7 GHz Intel i7-3820QM processor. The performance of the proposed hierarchical methodology for detection, classification, and location FDEs in power systems is evaluated through three case studies on the standard IEEE benchmark systems: IEEE 14 bus system, New England 39-bus system, and modified IEEE 118-bus system [40]. The historical database of FDEs in real power systems is not easily accessible to train the proposed models. Besides, it is crucial to consider a large number of scenarios for the intended purpose. Therefore, the required training dataset has been generated in DIgSILENT PowerFactory software environment. To analyze the network in various operating conditions, in each scenario, the entire network's load is changed, adopting the probabilistic distribution function of Eq. (16) [41], and then in each scenario, GT, LD, LO, and fault events are simulated. Adopting this probabilistic distribution function, all network loads' active and reactive power amounts are

modified to place the network at various operating points.

$$x_i = \mu_i + \sigma\sqrt{-2 \times ln(u_1)} \times cos(2\pi u_2) \quad (16)$$

where x_i is the amount of active or reactive power of load i. $\mu_i$and σ represent the base value of the active or reactive power of load i and standard deviation, respectively. $u_1$and u_2 are also two random numbers from a standard uniform distribution. σ is considered as 0.2 in this study.

For IEEE 14 bus system and New England 39-bus system, 2998 and 4028 scenarios are simulated, respectively, for both FDE classification and detection, which are summarized in Table 3. Almost 75% of the dataset is employed for model training, and the rest 25% for testing. As mentioned, the 39-bus network utilizes 5 PMUs to get the ROCOF time series. Also, in the 14-bus network, 5 PMUs installed according to Fig. 7 are used for the purpose. In the modified IEEE 118-bus system, PMUs are installed on all busbars that are connected to the generator (19 PMUs in total) due to the extensive system.

3.1. Numerical results of FDE classification and location

After fitting a DL model, it is crucial to evaluate its performance on the test dataset. The proposed models have been examined within recall, precision, and F1-score criteria in addition to accuracy.

***Recall*:** Recall criterion for classifying FDEs is expressed as the proportion of correctly classified FDEs of class*i*by the model (true positive or T_p) to the total number of class*i*events in the test database. The whole incorrectly categorized FDEs of class*i*are known as false negative (F_N). The high value for the recall metric means a small amount of class *i*data incorrectly classified in another class. Recall is calculated by

$$recall = \frac{T_P}{T_P + F_N} \quad (17)$$

***Precision*:** This criterion, calculated by Eq. (18), represents the ratio of correctly identified class*i*to the sum of T_p and the total number of incorrect FDEs assigned to class*i* (false positive or F_p). The high value of the precision criterion indicates the small amount of incorrectly categorized data in a particular class.

$$precision = \frac{T_P}{T_P + F_P} \quad (18)$$

***F1-score*:** Using both Recall and precision criteria, this criterion is obtained by Eq. (19), which draws a more accurate picture of how the classifier model operates on all classes in the dataset over the accuracy criterion.

$$F1 = 2\frac{recall \times precision}{recall + precision} = \frac{T_P}{T_P + 0.5(F_N + F_P)} \quad (19)$$

Table 4 demonstrates a comparison between RNN and LSTM proposed models in terms of accuracy, average recall, average precision, average F1-score, and convergence epochs for classification FDEs and GT, LO, LD, and fault location in three benchmark systems. As shown in Table 4, in the IEEE 14 bus system, both the suggested RNN and LSTM models categorize FDEs and locate GT, LO, and fault events with 100% accuracy. LD location is also detected with 99.47% and 99.40% accuracy by RNN and LSTM models, respectively. Although both models have a robust performance for identifying FDEs, the RNN method has a relatively higher convergence speed in GT location and fault location.

The New England 39-bus system's numerical results are also presented in Table 4, with the LSTM model classifying FDEs with 100% accuracy and the RNN model with 99.60%. Locating GT and LD events with the RNN model is done with 100% and 98.79% accuracy, respectively, which is more accurate than the 99.60% and 98.38% results of the LSTM model. Both models are challenged to locate LO and fault in the system. LO can cause cascading GT or LD events, increasing or decreasing the frequency/ROCOF, respectively. Also, with the occurrence of a short circuit on the transmission line, the generators' generating capacity increases, and on the other hand, due to this fault, the load demand connected to the end of the faulty line may not be provided; these also mean two different behaviors in the system frequency. So with the occurrence of LO and fault, different frequency behaviors can happen across the network. Also, the number of candidates in question as the event's location is high (34 points). With these interpretations, the models have difficulty understanding the exact location of LO and fault accurately, and there are more likely to overfit. However, both proposed models recognize the location of these two FDEs very strongly. The RNN model identifies LO and fault location with 96.37% and 93.15% accuracy, respectively, and these results in the LSTM model are 90.32% and 92.34%, respectively.

The performance of the proposed models in the 118-bus system is different from the other two systems studied. In this system, in all scenarios, the LSTM model has relatively better accuracy. However, in terms of convergence speed, the RNN model is still faster. The LSTM and RNN models identify the type of FDEs with 98.30% and 98.01% accuracy, respectively. Both models still detect the location of GT with very high accuracy (100%). Although they have pinpointed the location of LD and fault events with great accuracy, both models have encountered a little challenge in locating LO events in the system; the LSTM and RNN models were able to locate LO with 87.80% and 87.68% accuracy, respectively. The recall value for one of the lines is NAN, and both proposed models have not been able to locate any of the LO events of that line correctly. As a result, the average recall and F1-score for the LO classifier will be NAN.

According to the outcomes, it can be seen that the suggested RNN model for locating FDEs has a more reliable performance than the LSTM model in IEEE 14 bus system and New England 39-bus system. However, in the 118-bus system, the opposite is true. Also, overall, RNN model converges in a smaller number of epochs.

Also, the proximity of the F1-score to their accuracy for all scenarios confirms the efficiency of the models. Overall, the value of recall, precision, and consequently F1-score criteria in the proposed RNN model is higher than the proposed LSTM model.

Ref. [20] proposed two CNN-based deep learning models to detect the type of GT and LD events that simultaneously use the ROCOF signal and relative angle shift to train and test the models. In the paper, to collect data and use the CNN models, an unlimited number of measurements has been employed to record ROCOF and relative angle shift

Table 3
Distribution of FDEs into test and training datasets.

Event		IEEE 14 bus system			New England 39-bus system		
		Training set (75%)	Test set (25%)	Total	Training set (75%)	Test set (25%)	Total
FDE Classification	GT	566	184	750	756	244	1000
	LO	565	187	750	765	255	1020
	LD	569	179	748	723	265	988
	Fault	551	201	750	760	260	1020
FDE Location	GT	562	188	750	750	250	1000
	LO	564	188	750	765	255	1020
	LD	561	187	748	741	247	988
	Fault	564	188	750	765	255	1020

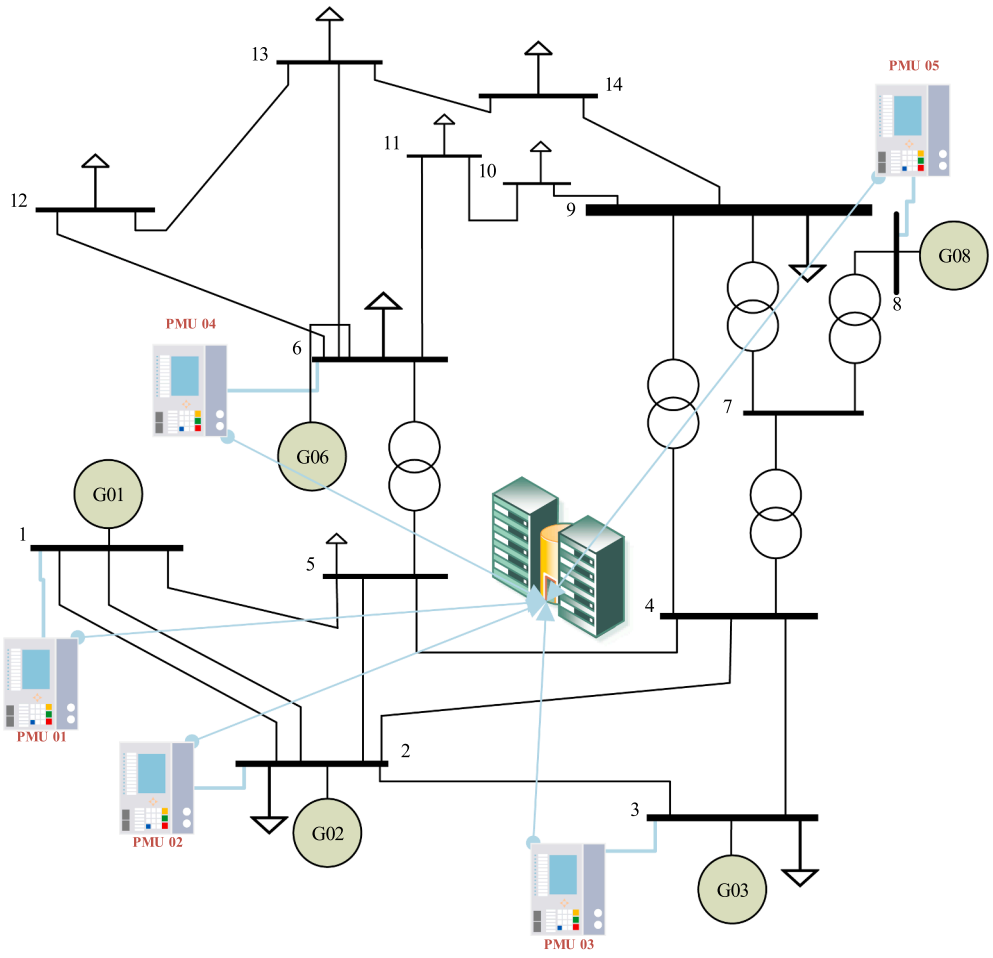


Fig. 7. Single-line diagram of the IEEE 14 bus test system.

Table 4
Comparison of results using evaluation criteria.

FDE Identification		Evaluation Criteria	IEEE 14 bus system		New England 39-bus system		Modified IEEE 118-bus system	
			Proposed RNN	Proposed LSTM	Proposed RNN	Proposed LSTM	Proposed RNN	Proposed LSTM
FDE Classification		Accuracy (%)	**100**	**100**	**99.60**	**100**	**98.01**	**98.30**
		Ave. Recall (%)	100	100	99.56	100	97.91	97.71
		Ave. Precision (%)	100	100	99.62	100	97.76	97.33
		Ave. F1-score (%)	100	100	99.61	100	97.84	97.52
		Conv. Epochs	20–30	20–30	60–80	80–100	30–40	30–60
FDE Location	GT	Accuracy (%)	**100**	**100**	**100**	**99.60**	**100**	**100**
		Ave. Recall (%)	100	100	100	99.66	100	100
		Ave. Precision (%)	100	100	100	99.52	100	100
		Ave. F1-score (%)	100	100	100	99.59	100	100
		Conv. Epochs	5–10	20–30	10–20	20–40	10–20	40–50
	LO	Accuracy (%)	**100**	**100**	**96.37**	**90.32**	**87.68**	**87.80**
		Ave. Recall (%)	100	100	96.94	90.87	87.29	88.33
		Ave. Precision (%)	100	100	96.58	91.56	NAN	NAN
		Ave. F1-score (%)	100	100	96.76	91.21	NAN	NAN
		Conv. Epochs	80–100	80–100	400–600	600–800	300–400	300–400
	LD	Accuracy (%)	**99.47**	**99.40**	**98.79**	**98.38**	**95.38**	**96.26**
		Ave. Recall (%)	99.62	98.76	98.22	97.81	95.76	95.11
		Ave. Precision (%)	99.59	98.30	98.36	98.74	95.43	96.30
		Ave. F1-score (%)	99.60	98.53	98.29	98.27	95.60	95.70
		Conv. Epochs	150–200	100–120	150–200	200–300	200–250	200–300
	Fault	Accuracy (%)	**100**	**100**	**93.15**	**92.34**	**95.25**	**95.82**
		Ave. Recall (%)	100	100	92.69	92.48	95.12	95.99
		Ave. Precision (%)	100	100	93.38	92.25	NAN	NAN
		Ave. F1-score (%)	100	100	93.03	92.37	NAN	NAN
		Conv. Epochs	30–50	60–100	800–1000	800–1000	400–600	400–600

signals. The first CNN model is based on color images, and the second CNN model is based on gray-scaled images. By training 75% of the available data, the accuracy of both models in detecting the type of GT events is 100%, but in detecting the type of LD events, they have reached 85.53% and 89.47%, respectively. Ref. [20] concentrates on identifying the type of FDEs and does not examine the location of them.

Ref. [42] proposed a model based on Maximum Likelihood Criterion (MLE) and Density-based spatial clustering of applications with noise (DBSCAN) for classifying and locating events based on changes in operating point in voltage, current, active power, reactive power, and frequency. Ref. [42] utilized 5 PMUs data to classify and locate events in the New England 39-bus system; nevertheless, it can only detect and locate events that occur near PMUs. However, it has not been entirely successful in detecting these events. Therefore, the model cannot identify and locate events that occur over the whole system.

Compared to the existing works, the models proposed in the paper, which use only ROCOF signal data from a limited number of PMUs, can identify and locate all GT, LD, LO, and three-phase faults that occur throughout the system with very competitive accuracy. The results are summarized in Table 4.

3.2. Training and test results of FDE classification and location

Fig. 8 illustrates the convergence speed, accuracy, and loss of training and validation data of proposed models for classification FDEs and locating them on the New England 39-bus system. In Fig. 8a, both training and validation accuracy converge after approximately the 40th epochs to classify FDEs in the LSTM model. Interestingly, in Fig. 8b and Fig. 8c, which are for GT and LD location, respectively, the LSTM model's convergence speed on test data is almost faster than the training data. This is because the ROCOF signal is received from the PMUs installed on the generators' busbars, so the model efficiently identifies the GT location wherever in the network. Also, discontinuing each of the loads in various operating situations of the system will result in a relatively similar frequency response for each load, which facilitates the detection of the exact LD location.

The LSTM model is lightly overfitted in locating LO in Fig. 8d, and the optimizer has not been able to update the model parameters optimally, and hence the quantity of validation loss has been associated with numerous fluctuations. Nevertheless, training loss is well converged. The reason for these fluctuations is the several frequency behaviors of individual LO in different network operating conditions. However, validation accuracy still converged after 600 epochs and the model detects the LO location with slightly higher accuracy of 90%. Although in the LSTM model, fault location is also associated with small overfitting, the validation data's accuracy has touched 92% of accuracy (Fig. 8e).

In contrast, the RNN model fits firmly on validation and training data to identify FDEs, and the convergence speed and accuracy of the model are higher than the LSTM model. In this model, similar to the LSTM model, the convergence rate on the classification of FDEs and the location of GT and LD on the validation data is higher than the training data (Fig. 8f, Fig. 8g, and Fig. 8h); For instance, in Fig. 8g, after only two epochs, the model accuracy of the validation data reaches 100%, but the training data converges around the 20th epoch. Fig. 8i shows the RNN model's performance for locating LO, which is more reliable than the LSTM model, and parameters are well adjusted. Similarly, the model has very little overfitted on fault location, but it is still more accurate on validation data over the LSTM model.

Although the LSTM model has shown excellent performance in various issues, due to its more complex structure than the RNN model and its high sensitivity to initial weighting and increasing the number of parameters, it is almost prone to overfitting in FDE detection problem. According to the outcomes, both proposed models have satisfying performance in identifying FDEs and locating them. Nevertheless, the performance of the RNN model seems to be slightly better than the LSTM

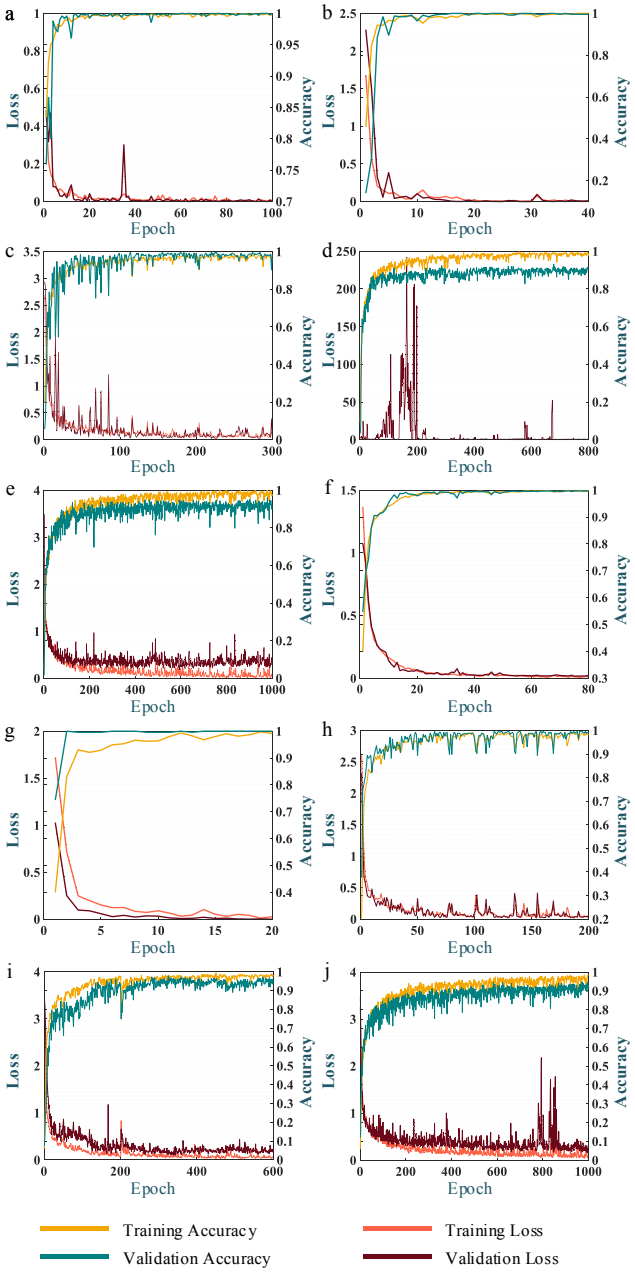


Fig. 8. Comparison between accuracy and loss of proposed models on training and test data in New England 39-bus system; (a) LSTM model for event classification; (b) LSTM model for GT location; (c) LSTM model for LD location; (d) LSTM model for LO location; (e) LSTM model for fault location; (f) RNN model for event classification; (g) RNN model for GT location; (h) RNN model for LD location; (i) RNN model for LO location; (j) RNN model for fault location.

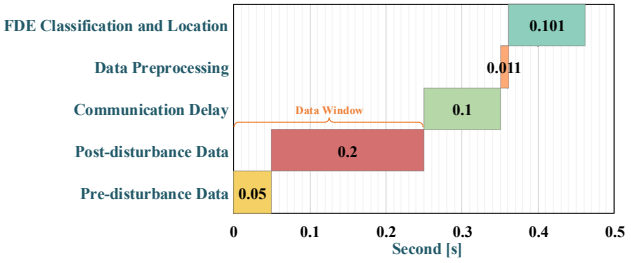


Fig. 9. Real-time process timeline.

model in IEEE 14 bus system and New England 39-bus system. Though, in the IEEE 118-bus system, the accuracy of the LSTM model in all scenarios is higher than the RNN model.

Fig. 9 demonstrates real-time estimation for classification and locating FDEs by the proposed RNN model, in which the data window contains 3 pre-disturbance samples and 12 post-disturbance samples. Also, 100 ms is considered for communication delay from PMUs to PDC. The proposed RNN model requires only 112 ms to classify and locate an FDE, including data preprocessing and run time. The proposed LSTM model also does this in 115 ms. A comparison between different models' run times to classify and locate FDEs is presented in Table 5. The color and gray-based CNN models presented in [20] only classify FDEs in 120 and 115 ms, respectively. The fastest performance is the model presented in [42], which classifies and locates events in about 90 ms, but the problem with this method is that it only can detect FDEs that occur near PMUs.

3.3. Comparison of locating FDEs employing the proposed hierarchical framework and the direct method

If the proposed hierarchical framework is not utilized to classify and locate FDEs, and the proposed RNN and LSTM models in Fig. 5 are used directly to locate FDEs (i.e., by omitting the event class identification stage), in this case, the models will reach the accuracy presented in Table 6. According to the results of Table 6, in IEEE 14 bus system, the RNN and LSTM models identify the FDEs' location with 98.80% and 98.40% accuracy employing the direct method, respectively. While applying the proposed hierarchical structure, with 100% accuracy, the class of FDE is detected by both models; besides, the location of all GT, LD, and faults events is identified with 100% accuracy, and the location of LO events with approximately 99.40% accuracy (Table 4). Therefore, the performance and accuracy of the proposed hierarchical models are better and more reliable than using the direct method to identify the location of FDEs. The same result is valid for the New England 39-bus system.

3.4. Performance in the absence of some PMUs information

In addition to the FDEs covered in this paper, cyber-attacks on power systems are among the latest and essential topics that can have a devastating effect on the security, reliability, and stability of power systems. It is crucial to adopt appropriate defense strategies to protect the system and data against cyber threats [43,44]. To this end, [45] has proposed a method named Deeply-Hidden Moving-Target-Defense (DH-MTD) that can deeply hide reactance changes in unbalanced AC distribution networks systems to defend against cyber-attacks. DH-MTD can hide self and mutual reactance parameters in the Reactance Perturbation (RP) matrix of transmission lines.

PMU measurements may be associated with erroneous data caused by (1) intentional failure of information in a cyber-attack and (2) inadvertent failure of data in the stages of communication, processing, storage [46]. In this paper, the performance of the proposed models in case ROCOF data is not received from one PMU, and 2 PMUs is investigated. The same pre-trained models are employed for this purpose. In the test data for every FDE, one or two rows of data are deleted randomly. The model performance is then evaluated for these new test data, illustrated in Fig. 10 and Fig. 11.

As shown in Fig. 10 in both test systems, if the information of one of the PMUs does not reach the network control center, the proposed models can still classify FDEs with reasonable accuracy. In the New England 39-bus system, the LSTM and RNN models categorize FDEs with 95.87% and 94.46% accuracy, respectively, and in IEEE 14 bus system with 83.07% and 81.60% accuracy, respectively. Nevertheless, the

Table 5
Time comparison for FDE classification and location.

Model	Proposed RNN	Proposed LSTM	CNN (Color) [20]	CNN (Gray-scale) [20]	DBSCAN & MLE [42]
Time (ms)	112	115	120*	115*	90

* Ref. [20] specifies only the type of FDEs and does not examine the location of them.

Table 6
Results of the direct method to locate FDEs (without using the hierarchical method proposed in the paper).

FDE Location	IEEE 14 bus system		New England 39-bus system	
Evaluation criteria	Proposed RNN	Proposed LSTM	Proposed RNN	Proposed LSTM
Accuracy (%)	98.80	98.40	90.52	92.24
Conv. Epochs	100–150	100–150	400–500	400–500

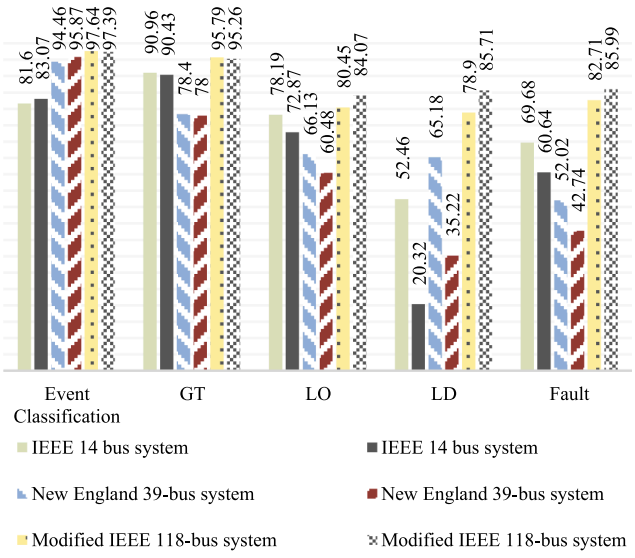


Fig. 10. Performance of models in the absence of information of one of the PMUs in term of accuracy (%).

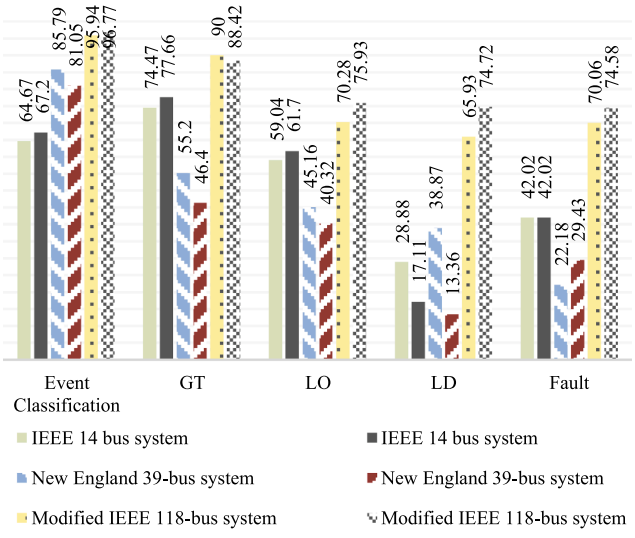


Fig. 11. Performance of models in the absence of information of two of the PMUs in term of accuracy (%).

models are challenged to detect the FDEs location; the best accuracy is to locate GT in both systems. The LSTM model has performed poorly in both systems to detect LD location, while the RNN model's accuracy for this purpose is moderately better. Also, the RNN model is more accurate in detecting LO and fault location than the other model.

As illustrated in Fig. 11, if out of a whole 5 PMUs used in test systems, the ROCOF time series of 2 PMUs is not collected by the PDC, the proposed RNN and LSTM models still perform well in detecting the FDE type in the New England 39-bus system, i.e., 85.79% and 81.05%, respectively. However, the accuracy of the models for locating FDEs has been drastically decreased. This inaccuracy of models is due to the tiny amount of data that has been used as model inputs from the beginning. In the absence of data received from 2 PMUs in the New England 39-bus system, the RNN model identifies the locations of the GT, LO, and LD with 55.2%, 45.16%, and 38.87% accuracy, respectively, which is performed better over the results of the LSTM method. In the IEEE 14 bus system, FDEs were detected with 64.2% and 67.2% accuracy using RNN and LSTM models. Similarly, models in this network also face serious challenges in locating events if they do not have access to the data of 2 PMUs. However, the proposed RNN and LSTM models identified the GT event locations with relatively satisfactory accuracy of 74.77% and 77.66%, respectively.

Due to the large of the IEEE 118-bus system, 19 PMUs have been installed from the beginning in the system. Hence, if the information about one or two of the PMUs of the system does not reach the model, the model will not face many difficulties and can categorize and locate FDEs with great accuracy in both RNN and LSTM models. Although the accuracy of both models is close to each other, the LSTM model performs much better in detecting the location of LO and LD events.

Although the proposed models performed relatively well in identifying the type of FDEs, they were less accurate in detecting FDEs location. Therefore, for future work, the authors are considering a method based on artificial intelligence algorithms to apply correct data instead of lost information from PMUs and detect the event of cyber-attacks with intelligent algorithms.

4. Conclusions

Real-time detection and location of FDEs in power systems are crucial in ensuring the system's reliability and stability operation, taking appropriate action, and reducing financial losses. Compared to other similar literature, which concentrated only on classifying or locating events separately, this paper developed a comprehensive hierarchical framework for real-time detection, classification, and location of FDEs. It employs a limited data window for the purpose, which reducing storage problems and data processing time. Two RNN and LSTM models have been developed and trained to evaluate, categorize and locate each of the GT, LO, LD, and fault events, and the results have been evaluated using the accuracy, recall, precision, and F1-score criteria. The performance of two models on modified IEEE 118-bus system, New England 39-bus and IEEE 14 bus systems were validated. According to the numerical results, both proposed models can categorize and locate FDEs with much higher accuracy than related works. The models' performance was also assessed if data from one or two PMUs were not obtained; both models could operate satisfactorily in classifying FDEs but were challenged in locating them because a limited number of PMUs were employed at the beginning. Overall, according to the outcomes, it was affirmed that the proposed RNN model works more trustworthy over the LSTM model in different scenarios and, in addition to relatively higher accuracy, also converges in a smaller number of epochs.

For future works, the authors seek to detect and locate the simultaneous occurrence of several FDEs in the system employing advanced deep learning models. The authors are also developing an artificial intelligence model for identifying cyber-attacks to replace lost or defective information with correct information.

CRediT authorship contribution statement

Mohammad Reza Shadi: Conceptualization, Methodology, Software, Formal analysis, Data curation, Writing - original draft, Visualization. **Mohammad-Taghi Ameli:** Conceptualization, Supervision, Project administration, Funding acquisition, Writing - review & editing. **Sasan Azad:** Conceptualization, Methodology, Software, Validation, Writing - review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

[1] Abdulraheem BS, Gan CK. Power system frequency stability and control: Survey. Int J Appl Eng Res 2016;11:5688–95.
[2] Bank JN, Gardner RM, Wang JK, Arana AJ, Liu Y. Generator trip identification using wide-area measurements and historical data analysis. In: 2006 IEEE PES power syst conf expo PSCE 2006 – Proc 2006:1677–81. https://doi.org/10.1109/PSCE.2006.296165.
[3] Morales JA, Orduña E, Rehtanz C, Cabral RJ, Bretas AS. Ultra high speed deterministic algorithm for transmission lines disturbance identification based on principal component analysis and Euclidean norm. Int J Electr Power Energy Syst 2016;80:312–24. https://doi.org/10.1016/j.ijepes.2016.01.043.
[4] Miranda V, A. Cardoso P, Bessa RJ, Decker I. Through the looking glass: seeing events in power systems dynamics. Int J Electr Power Energy Syst 2019;106:411–9. https://doi.org/10.1016/j.ijepes.2018.10.024.
[5] Electricity Consumers Resource Concil (ELCON). The economic impacts of the August 2003 blackout. New York; 2004. p. 1–10. Available: https://elcon.org/wp-content/uploads/Economic20Impacts20of20August20200320Blackout1.pdf.
[6] CIGRE. Analysis and modeling needs of power systems under major frequency disturbances. CIGRE Task Force 380214; 1999. p. 39–48. Available: https://books.google.com/books/about/Analysis_and_Modelling_Needs_of_Power_Sy.html?id=4OSatgAACAAJ.
[7] Veloza OP, Santamaria F. Analysis of major blackouts from 2003 to 2015: classification of incidents and review of main causes. Electr J 2016;29(7):42–9. https://doi.org/10.1016/j.tej.2016.08.006.
[8] Anagnostatos SD, Halevidis CD, Polykrati AD, Bourkas PD, Karagiannopoulos CG. Examination of the 2006 blackout in Kefallonia Island, Greece. Int J Electr Power Energy Syst 2013;49:122–7. https://doi.org/10.1016/j.ijepes.2012.12.003.
[9] Frigo G, Derviskadic A, Zuo Y, Paolone M. PMU-based rocof measurements: uncertainty limits and metrological significance in power system applications. IEEE Trans Instrum Meas 2019;68(10):3810–22. https://doi.org/10.1109/TIM.1910.1109/TIM.2019.2907756.
[10] Khaleghi A, Sadegh MO, Ahsaee MG. Permanent fault location in distribution system using phasor measurement units (PMU) in phase domain. 2709-2709 Int J Electr Comput Eng 2018;8(5). https://doi.org/10.11591/ijece:v8i5.pp2709-2720.
[11] Mirshekali H, Dashti R, Keshavarz A, Torabi AJ, Shaker HR. A novel fault location methodology for smart distribution networks. IEEE Trans Smart Grid 2021;12(2):1277–88. https://doi.org/10.1109/TSG.516541110.1109/TSG.2020.3031400.
[12] Thilakarathne C, Meegahapola L, Fernando N. Real-time voltage stability assessment using phasor measurement units: influence of synchrophasor estimation algorithms. Int J Electr Power Energy Syst 2020;119:105933. https://doi.org/10.1016/j.ijepes.2020.105933.
[13] Cui M, Wang J, Tan J, Florita AR, Zhang Y. A novel event detection method using PMU data with high precision. IEEE Trans Power Syst 2019;34(1):454–66. https://doi.org/10.1109/TPWRS.2018.2859323.
[14] Ma J, Zhang Pu, Fu H-J, Bo Bo, Dong Z-Y. Application of phasor measurement unit on locating disturbance source for low-frequency oscillation. IEEE Trans Smart Grid 2010;1(3):340–6. https://doi.org/10.1109/TSG.2010.2071889.
[15] Sobrinho ASF, Flauzino RA, Liboni LHB, Costa ECM. Proposal of a fuzzy-based PMU for detection and classification of disturbances in power distribution networks. Int J Electr Power Energy Syst 2018;94:27–40. https://doi.org/10.1016/j.ijepes.2017.06.023.
[16] Duan N, Stewart EM. Frequency event categorization in power distribution systems using micro PMU measurements. IEEE Trans Smart Grid 2020;11(4):3043–53. https://doi.org/10.1109/TSG.2020.2967641.
[17] Kim D-I, White A, Shin Y-J. PMU-Based event localization technique for wide-area power system. IEEE Trans Power Syst 2018;33(6):5875–83. https://doi.org/10.1109/TPWRS.5910.1109/TPWRS.2018.2824851.
[18] Khodayar M, Liu G, Wang J, Khodayar ME. Deep learning in power systems research: a review. CSEE J Power Energy Syst 2021;7:209–20. https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=9265470.
[19] Dabbaghjamanesh M, Moeini A, Kavousi-Fard A. Reinforcement learning-based load forecasting of electric vehicle charging station using Q-learning technique. IEEE Trans Ind Informatics 2021;17(6):4229–37. https://doi.org/10.1109/TII.942410.1109/TII.2020.2990397.

[20] Wang W, Yin He, Chen C, Till A, Yao W, Deng X, et al. Frequency disturbance event detection based on synchrophasors and deep learning. IEEE Trans Smart Grid 2020; 11(4):3593–605. https://doi.org/10.1109/TSG.516541110.1109/TSG.2020.2971909.
[21] Chadha GS, Panambilly A, Schwung A, Ding SX. Bidirectional deep recurrent neural networks for process fault classification. ISA Trans 2020;106:330–42. https://doi.org/10.1016/j.isatra.2020.07.011.
[22] Dehghani M, Kavousi-Fard A, Dabbaghjamanesh M, Avatefipour O. Deep learning based method for false data injection attack detection in AC smart islands. IET Gener Transm Distrib 2020;14(24):5756–65. https://doi.org/10.1049/gtd2.v14.2410.1049/iet-gtd.2020.0391.
[23] Zhao H, Sun S, Jin B. Sequential fault diagnosis based on LSTM neural network. IEEE Access 2018;6:12929–39. https://doi.org/10.1109/ACCESS.2018.2794765.
[24] Appiah AY, Zhang X, Ayawli BBK, Kyeremeh F. Long short-Term memory networks based automatic feature extraction for photovoltaic array fault diagnosis. IEEE Access 2019;7:30089–101. https://doi.org/10.1109/ACCESS.2019.2902949.
[25] Alam M, Kundu S, Thakur SS, Banerjee S. PMU based line outage identification using comparison of current phasor measurement technique. Int J Electr Power Energy Syst 2020;115:105501. https://doi.org/10.1016/j.ijepes.2019.105501.
[26] Jena MK, Panigrahi BK, Samantaray SR. Online detection of tripped transmission line to improve wide-area SA in power transmission system. IET Gener Transm Distrib 2018;12(2):288–94. https://doi.org/10.1049/gtd2.v12.210.1049/iet-gtd.2016.1964.
[27] Bi TS, Song XN, Wu JT, Yang QX. Novel method for disturbance identification in power systems. In: 2006 IEEE power eng soc gen meet PES; 2006. https://doi.org/10.1109/pes.2006.1709431.
[28] Song Y, Wang W, Zhang Z, Qi H, Liu Y. Multiple event detection and recognition for large-scale power systems through cluster-based sparse coding. IEEE Trans Power Syst 2017;32(6):4199–210. https://doi.org/10.1109/TPWRS.2017.2656146.
[29] Azizi S, Rezaei Jegarluei M, Dobakhshari AS, Liu G, Terzija V. Wide-area identification of the size and location of loss of generation events by sparse PMUs. IEEE Trans Power Deliv 2020;8977:1–10. https://doi.org/10.1109/TPWRD.2020.3047228.
[30] Xia T, Zhang H, Gardner R, Bank J, Dong J, Zuo J, et al. Wide-area frequency based event location estimation. IEEE Power Eng Soc Gen Meet PES 2007;2007:1–7. https://doi.org/10.1109/PES.2007.386018.
[31] Dutta Chowdhury S. A novel data-driven method for detection and localization of power system events causing violation of pre-defined ROCOF limits. Electr Power Syst Res 2021;192:106895. https://doi.org/10.1016/j.epsr.2020.106895.
[32] Machowski J, Bialek JW, Bumby JR. Power system dynamics: stability and control. John Wiley and Sons Ltd.; 2008.
[33] Jagtap AD, Kawaguchi K, Karniadakis GE. Adaptive activation functions accelerate convergence in deep and physics-informed neural networks. J Comput Phys 2020; 404:109136. https://doi.org/10.1016/j.jcp.2019.109136.
[34] Wu S, Li G, Deng L, Liu L, Wu D, Xie Y, et al. L1 -norm batch normalization for efficient training of deep neural networks. IEEE Trans Neural Networks Learn Syst 2019;30(7):2043–51. https://doi.org/10.1109/TNNLS.596238510.1109/TNNLS.2018.2876179.
[35] Wang S, Chen H. A novel deep learning method for the classification of power quality disturbances using deep convolutional neural network. Appl Energy 2019; 235:1126–40. https://doi.org/10.1016/j.apenergy.2018.09.160.
[36] Srivastava N, Hinton G, Krizhevsky A, Sutskever I, Salakhutdinov R. Dropout: a simple way to prevent neural networks from overfitting. J Mach Learn Res 2014; 15:1929–58.
[37] Yadav R, Pradhan AK, Kamwa I. Real-time multiple event detection and classification in power system using signal energy transformations. IEEE Trans Ind Inform 2019;15(3):1521–31. https://doi.org/10.1109/TII.2018.2855428.
[38] Wen L, Li X, Gao L. A new two-level hierarchical diagnosis network based on convolutional neural network. IEEE Trans Instrum Meas 2020;69(2):330–8. https://doi.org/10.1109/TIM.1910.1109/TIM.2019.2896370.
[39] Zhu L, Hill DJ, Lu C. Hierarchical deep learning machine for power system online transient stability prediction. IEEE Trans Power Syst 2020;35(3):2399–411. https://doi.org/10.1109/TPWRS.5910.1109/TPWRS.2019.2957377.
[40] Canizares C, Fernandes T, Geraldi E, Gerin-Lajoie L, Gibbard M, Hiskens I, et al. Benchmark models for the analysis and control of small-signal oscillatory dynamics in power systems. IEEE Trans Power Syst 2017;32(1):715–22. https://doi.org/10.1109/TPWRS.5910.1109/TPWRS.2016.2561263.
[41] Anders GJ. Probability concepts in electric power systems. New York: John Wiley and Sons; 1990.
[42] Pandey S, Srivastava AK, Amidan BG. A real time event detection, classification and localization using synchrophasor data. IEEE Trans Power Syst 2020;35(6): 4421–31. https://doi.org/10.1109/TPWRS.2020.2986019.
[43] Angle MG, Madnick S, Kirtley JL, Khan S. Identifying and anticipating cyberattacks that could cause physical damage to industrial control systems. IEEE Power Energy Technol Syst J 2019;6(4):172–82. https://doi.org/10.1109/JPETS.2019.2923970.
[44] Chen C, Cui M, Fang X, Ren B, Chen Y. Load altering attack-tolerant defense strategy for load frequency control system. Appl Energy 2020;280:116015. https://doi.org/10.1016/j.apenergy.2020.116015.
[45] Cui M, Wang J. Deeply hidden moving-target-defense for cybersecure unbalanced distribution systems considering voltage stability. IEEE Trans Power Syst 2021;36: 1961–72. https://doi.org/10.1109/TPWRS.2020.3031256.
[46] Khare G, Mohapatra A, Singh SN. A real-time approach for detection and correction of false data in PMU measurements. Electr Power Syst Res 2021;191:106866. https://doi.org/10.1016/j.epsr.2020.106866.